

Question ID: 0aaef7aa

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

The function p is defined by $p(n) = 7n^3$. What is the value of n when $p(n)$ is equal to 56?

Answer

- A. 2
- B. $\frac{8}{3}$
- C. 7
- D. 8

Correct Answer: A

Rationale

Choice A is correct. It's given that $p(n) = 7n^3$. Substituting 56 for $p(n)$ in this equation yields $56 = 7n^3$. Dividing each side of this equation by 7 yields $8 = n^3$. Taking the cube root of each side of this equation yields $2 = n$. Therefore, when $p(n)$ is equal to 56, the value of n is 2.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.